# Considerations When Validating Your Analyst® Software Per GAMP® 5

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## INTRODUCTION

The purpose of this white paper is to assist users in understanding what must be addressed when validating their Analyst? Software in accordance with GAMP® 5.

### What is Software Validation?

Software validation, in the context of government regulation, is distinct from hardware qualification (ie. IQ/OQ/IPV).[1]

Rules and regulations for laboratory systems can be found in 21 CFR 211 (Good Manufacturing Practice), 21 CFR 58 (Good Laboratory Practice), 21 CFR 820 (Quality System Regulation for Medical Devices), 21 CFR Part 11 (Electronic Records and Electronic Signatures), as well as several others.[2,3] An analysis of the relevant regulations is beyond the scope of this paper. However, the spirit of the regulations can be summarized as follows:

Validation is a required activity that should document that the system is secure, reliable, and fit for its intended purpose. It must also provide procedures so the system remains in a validated state throughout its operation.

The FDA considers software validation to be "confirmation by examination and provision of objective evidence that software specifications conform to user needs and intended uses, and that the particular requirements implemented through software can be consistently fulfilled."[4] Note the terms "objective evidence", and "particular requirements".

Confirmation of conformity to user needs and intended uses is obtained by comparing actual system performance against particular or pre-determined requirements. During validation this comparison is accomplished by executing test procedures and collecting objective evidence (screen shots, printed reports, result data files, etc.). The point is not to produce a mountain of documentation, but rather to demonstrate that the software satisfies the intended use. It is also important to demonstrate that validation activities were properly planned and that the tests were executed according to the plan.

### Who is Responsible for Validation?

Under OECD5, validation is the responsibility of the "Test Site Management". In GLP, is it the responsibility of the "System Owner" or "Business Process Owner". Often, but not always, this is the GLP laboratory manager. While the laboratory manager will be a key member of the validation team, the team must include representatives from all validation stakeholders. The Quality Assurance team certainly has a role to play in validation. Ultimately upper management must provide the impetus and resources for validation. They must also "accept" that the system is validated.

While an organization can utilize third parties to design and perform the validation, the responsibility for validation and the maintenance of a validated state cannot be delegated.

### To Validate or Not?

The most important update in GAMP® 5 is the focus on risk management as it relates to patient safety.[4] GAMP® 5 requires validation if there could be an impact on patient safety, product quality, or data integrity.[6] Therefore, the decision to validate, what to validate and how to validate is largely an exercise in risk management.

When doing risk assessment in accordance with GAMP® 5, it is important to assess the probability and severity of harm to the patient if a fault occurs, rather than the probability of the fault occurring. Therefore, risk must assessed based on critical functionality, for example, in Analyst® acquisition of samples would be more critical than formatting reports. Therefore, more in depth testing would be needed for acquisitions.

GAMP® 5 also recognizes that higher system complexity increases the likelihood of risk.1 For example, a system configured to use Analyst Administrator Console (AAC) AAC for centralized security and project file management would be more complex than a stand alone system using Analyst®. Therefore, additional tests would be required to validate the system with AAC.

## The Cost of Compliance vs. The Cost of Non-compliance

The decision to forego validation when it is required amounts to the acceptance of the unmanageable risk of noncompliance. A brief review of recent judgments against pharmaceutical companies and independent/contract labs reveals that the cost of compliance may be thousands of dollars. But the cost of noncompliance can be millions of dollars along with lost revenue, lost productivity, possible process rework, and damaged investor and customer confidence and goodwill.

Compliance can be accomplished by validating critical sub-systems thoroughly and other functions minimally. This does not eliminate risk, but reduces it to manageable levels, while still controlling validation effort and expense. The decision to fully validate all components of a system further eliminates risk, but with a higher cost., not necessarily equivalent to the reduced risk.

## Prospective vs. Retrospective Validation

Ideally, the validation process should begin at the earliest stages of system procurement. It is not possible to wisely select a computerized system without an explicate understanding of the needs of stakeholders and the requirements of regulators. If one is to ensure that a system meets the needs it is intended to, then one must have a clear definition of what those requirements are.

In practice, it is sometimes necessary to validate an existing system already in use. In this case, it is still necessary to clearly state the requirements of the system, and to verify that those requirements are met.

The validation process continues throughout the lifecycle of the system. As changes to the system are made it will be necessary to confirm that the system remains validated in the face of those changes.

Planning for the ultimate retirement of the system is also part of the validation process.

## Controls: Technical and Procedural

To satisfy the regulations, it is necessary to place controls in the system. Controls can be of two types: technical controls and procedural controls. Technical controls are those controls that are enforced through hardware and software. They reduce human effort through automation thereby reducing the incidence of human error.

Procedural controls are processes that are documented, approved and enforced typically in a Standard Operating Procedure (SOP).

An example with both technical and procedure controls can be a door to a lab with an electronic lock. Procedurally, the lab should have an SOP describing the assignment, distribution, and maintenance of identification devices such as entry of a pass code, presentation of an electronic identification such as a badge, or use biometric identity verification. The badge or the biometric identity lock are technical controls because the lock uses hardware and software to allow or deny entry to the lab.

A procedural control would instruct a user to "identify" themselves to the lock in order to open the door. The badge must remain in the sole possession of the employee to whom it was assigned. If an employee lends her badge to another employee, who then uses it to access a restricted area, the procedural control is compromised.

As shown in this example, it is important to put both technical and procedural controls in place.

## Standard Operating Procedures

As described above SOPs are a major part of the procedural controls of the system. Some requirements, such as training, cannot be satisfied through technical controls, but must be satisfied through procedural controls. Some important SOPs are issuance and control of usernames and passwords, training procedures, change control procedures, documentation maintenance procedures, backup and restoration of data, and archival and retrieval of data. It is also worthwhile to document your company's software validation procedures in an SOP.

## Change Control

Change is inherent in any computerized system. As new requirements are identified, errors found, and procedures revised, changes to the system will be necessary. It is essential that changes to a validated system be carefully controlled. Any change contemplated should be documented, analyzed, and tested. It is not adequate to test only the change. A change to one subsystem might affect other, seemingly unrelated, parts of the system. Minimally, a change should be requested in writing via a Change Request. The Change Request must be analyzed and approved by the technical resources involved. The risk assessment may need to be updated. Finally the Change Request must by approved by the Quality Assurance Unit. The Change Control policy should be documented in a SOP.

Failure to control change will result in a system that is no longer validated, and will expose the business to the risk of non-compliance.

## Software Categories

Originally the GAMP® Good Practice Guide: Validation of Laboratory Computerized Systems classified computer software in five categories[3]. There were some changes to categorization of software in GAMP® 5 and category 2 was discontinued. The categories were not renumbered. Therefore, Analyst® remains in Category IV - configured Commercial off-the-shelf (Configurable COTS).[7]

Analyst® is configurable because it accommodates the storage and persistence of user names, passwords, customized audit trails and instrument configuration. The effort required to validate a configurable system such as Analyst®, is greater than that required to validate operating systems, firmware, and standard software.

Analyst® can also be customized using Microsoft Visual Basic (VB). Custom or bespoke software requires even greater validation effort.

## GAMP® 5 Increased Awareness of Configurable and Networked Systems

GAMP® 5 recognizes that most computerized systems are now based on configurable packages that utilize computer networks. Therefore, it recommends that software testing should focus on the specific configuration of the program rather than core operational characteristics, especially when the supplier can demonstrate testing of the core operational functionality of the system.[8] Therefore, Supplier Audit Programs have more importance in GAMP® 5 because more and more supplier certificates are accepted in lieu of actual supplier audits.

## Changes to the "V" Validation Life Cycle

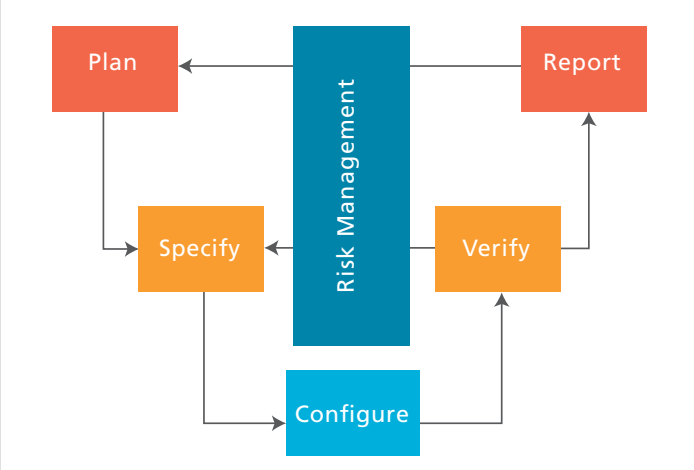



Figure 1. GAMP® 5 Validation Lifecycle[1]

Because GAMP® 5 recognizes that most systems are configurable software, it suggests a simplified "V" validation life cycle as shown in Figure 1 GAMP® 5 Validation Lifecycle.

## Important Documents

At a minimum, the validation documentation set should contain documents 01-08, 10 and 13 listed in the table below. Documents 09, 11 and 12 are special documents provided by AB SCIEX for software validation of Analyst?.

This table also includes a mapping of these documents to the GAMP® 5 Validation Lifecycle.

| Validation Document | Lifecycle Category |
|---|---|
| 01 Validation Plan (VP) | Plan |
| 02 Validation Risk Assessment (VRA) | Plan and Risk Management |
| 03 User Functional Requirements Specification (UFRS) | Specify |
| 04 System Design Specification (SDS) | Specify |
| 05 Test Plan | Plan |
| 06 Installation Qualification Protocol | Configure and Verify |
| 07 Operational Qualification Protocol | Verify |
| 08 Performance Qualification Protocol | Verify |
| 09 21 CFR Part 11 | Verify |
| 10 Traceability Matrix | Plan |
| 11 QAU Review | Verify/Report |
| 12 Vendor Assessment | Verify/Report |
| 13 Validation Summary Report (VSR) | Report |

It is important for the validation document set to be well organized. This can be accomplished by numbering the documents in a clear, easy to read manner. For example, each document is numbered 01-13 as shown above. Each document also has a code corresponding to the company, document and version such as CMPY-SDS-01 in the following example which is the typical introduction section in the Analyst? document set.

### 1. Introduction

Terms used in this document:

| | |
|---|---|
| **VENDOR** | AB SCIEX, Inc |
| **COMPANY** | COMPANY NAME (CMPY) |
| **TEAM** | Analyst® Software Validation Project Team defined in the Software Validation Plan CMPY-SVP-01 |
| **SYSTEM** | Defined in the System Design Specification CMPY-SDS-01 |
| **Analyst® IQ** | Analyst® Software Installation Qualification created by the TEAM in accordance with the Software Validation Plan CMPY-SVP-01 |

This type of cross-referencing makes for ease of use, modularity, and can assist any auditor in finding information quickly.

The following sections provide an overview of each validation document.

### The Validation Plan (VP)

The Validation Plan is a key strategic planning document[9] that describes the entire validation effort, and covers the system life cycle from inception to retirement. The regulations place great emphasis on the Validation Plan, because it is the key to controlling the validation project.

At a minimum, the Validation Plan should describe the scope of the validation project, the work to be done, the schedule of activities, and the individuals responsible for planning, execution, testing, and approval.

Additionally, instructions for testing including protocol execution and collection of objective evidence, as well as, post validation activities, deliverables, and instructions for identifying and documenting anomalies may be included in the Validation Plan or the Test Plan depending upon the needs of the System Owner.

### Validation Risk Assessment (VRA)

The VRA documents the risks in accordance with GAMP® 5 and includes prescribed mitigations for each risk.

### User Functional Requirements Specification (UFRS)

The UFRS contains objectively stated requirements that the system is intended to fulfill. It must address Analyst® Technical Controls, Procedural Controls, capacities, accuracy, security, fault tolerance, physical environment, and training requirements, among others. It is critical that the UFRS be a complete statement of the needs and objectives of the acquiring organization. A typical UFRS will contain up to several hundred unique requirements.

### The System Design Specification (SDS)

Because Analyst® software is configurable, it is adaptable to variations in instrument and peripheral equipment setup, security, and data processing. Therefore, it is necessary to describe the intended configuration in a System Design Specification (SDS). Some examples of the Analyst® features that are addressed in the SDS are: security and user roles, audit trail settings, equipment configuration, and quantitation settings.

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### Test Plan (TP)

The Test Plan ensures that each test is supported by a user requirement. At a minimum, it serves as a forward-pointing traceability matrix showing the relationship of the tests to the user requirements.

### Qualification Protocols

The qualification of the Analyst® software can be separated into four phases: Design Qualification (DQ), Installation Qualification (IQ), Operational Qualification (OQ), and Performance Qualification (PQ). Since it is not always clear in which phase a particular requirement or test belongs, the following guidance may be helpful.

Since Analyst® is a GAMP® 5 category IV software, the DQ is performed by AB SCIEX, the vendor. This can be verified by performing a vendor audit or accepting vendor certifications. AB SCIEX provides a standard postal audit that addresses the common elements of a vendor audit. It identifies the development and verification methodology and the quality procedures, such as ISO 9001 as implemented by AB SCIEX.

IQ, OQ and PQ testing involves the execution of a pre-defined set of tests. A test script contains the instruction, the expected result and the acceptance criteria. It also has a place for the tester to indicate whether the test passed or failed.

Good test scripts will reference each applicable requirement specifically. A single "test script or a certain number of tests might address one or several requirements. Normally, test scripts are organized around a specific range of functionality, such as security or data acquisition.

Test scripts must be carefully designed, and should include both positive and negative tests. For example, a test for password acceptance will include procedures to verify the result of entering a valid password, as well as the result of entering an invalid password.

If a test step fails, then an Anomaly Report is required. The deviation report should identify the nature of the deviation, the test script or procedure where the deviation occurred, proposed corrective action, and responsibility for implementation and verification.

### 21 CFR Part 11 Assessment

The Analyst® software can be configured to meet the requirements of 21 CFR Part 11, Electronic Records; Electronic Signatures (Part 11). Part 11 regulates the security, reliability and integrity of laboratory data, and the security and integrity of electronic signatures. The predicate rules contain relatively few signature requirements. Where signatures are required, such as in a data audit trail, Part 11 defines how an electronic signature must be derived and the meaning of the electronic signature. Many Part 11 requirements will be met with a combination of technical and procedural controls.

### The Analyst® software is not compliant with part 11 until the Analyst® software has been configured correctly and appropriate SOPs are in place.

The 21 CFR Part 11 Assessment contains a checklist to verify that 21 CFR Part 11 regulations have been met.

### Traceability Matrix™

The Traceability Matrix™ shows the relationship between each user requirement to a corresponding test script (in the case of technical controls) or SOP (for procedural controls). The TM makes it possible to confirm that each user requirement has been addressed and satisfied by the validation procedure.

### Quality Assurance Review

The Quality Assurance (QA) or Quality Control (QC) department must be actively engaged in the validation effort. Management's approval of validation depends on the recommendation of QA.

Fortunately, GAMP® 5 simplified the document approval process. Technical experts are now empowered to approve technical documentation. QA must ensure documents meet applicable regulations. For example QA should review a UFRS against the applicable regulations but the UFRS technical review is the responsibility of technical subject matter experts. Thus, QA no longer needs to sign a design specification because they can rely upon technical subject matter experts.

QA should verify that design specifications are being produced for projects (i.e. verify that processes are being followed) but QA does not need to sign every document in a project.[1]

After a final review for completeness, the QA department must submit recommendations to management regarding the release of the system for use. The best way to ensure that the QA department can recommend the release of the system is to include them in the validation effort throughout the lifecycle.

The Quality Assurance Review document provides a checklist to ensure the above criteria have been met. It also requires signatures denoting the pass/fail status of all validation documents.

### Vendor Assessment

The vendor assessment contains the standard postal audit for AB SCIEX.

### Validation Summary Report (VSR)

The VSR contains the results of the software validation project including the decision as to whether the system passed or failed.

### Planning for Maintenance of the Validated State

Analyst® validation is not a one-off process. The validation effort should encompass the entire system lifecycle, from inception to retirement. The most important tool for maintaining a system in its validated state is the Change Control Procedure. By carefully following a pre-defined plan for evaluating and approving changes to the system, the physical environment, and the procedural environment, a system can be maintained in a validated state over time.

### Replicate System Validation

When more than one instrument is being installed in a laboratory at the same time, the system can be replicated. When this occurs, it would be redundant and costly to perform a complete software validation on each system. The following provides guidance for tailoring the software validation for replicated systems.

First, the Validation Plan must describe that several instruments are being validated at the same time. The strategy is to test all requirements on one system called "First in Family" then test a subset of requirements on the replicated systems.

The Test Plan must denote tests to be performed on the First in Family and tests to be performed on Replicated Systems. Tests to be performed on Replicated Systems include security, audit trail configuration and acquisition settings. Thus, testing is limited to confirming that the configuration is correct. Additionally, an acquisition of a small number of samples (about 10) must be run to ensure the instrument is acquiring properly. Quantitation validation would not be necessary, nor testing reporting because these functions have been tested on the First in Family.

There should be two sets of IQ/OQ/PQ protocols. One for the First in Family, one for replicate systems.

The Validation Traceability Matrix should trace both the First in Family and the Replicated Systems. More than one trace table may be required.

The Validation Summary Report must list the validation status of each system and any anomalies encountered for each system.

## CONCLUSION

Analyst® validation need not be an onerous undertaking. By adopting the best practices prescribed by GAMP® 5, other regulatory bodies and professional societies, validation can be performed efficiently.

GAMP® 5 brought some changes to software validation. These include:

- Validation based on risk management with more testing required for functionality that could impact on patient safety, product quality, or data integrity.
- Increased Awareness of Configurable and Networked Systems
- Changes to the "V" Validation Life Cycle
- Simplified the document approval process

In addition to regulatory compliance, the processes and business objectives of the organization are enhanced by proper validation.

## CONTACT US

Contact your local AB SCIEX sales representatives or email AB SCIEX Validation Services at:
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